# TENCON 2026 Authors' Responses to Paper Review Comments

**Paper ID:** 1571282911 **Paper Title:** CA-CRS+: A Zone-Aware, Resource-Informed Crowd Risk Scoring Framework with Cause-Specific Gate Intervention Recommendation for Stampede Prevention
Transfer the tables below into the official TENCON response `.doc`. Page/paragraph numbers in the "Location" column are given as section references and finalized once the camera-ready is typeset (PDFeXpress). We thank both reviewers — every point below is addressed with newly added, reproducible experiments on public benchmarks (PETS2009 S2.L1 with camera calibration and MOT ground truth; ShanghaiTech A/B), and the evaluation was rebuilt end-to-end.

## Reviewer 1

| # | Reviewer's comments | Authors' response | Location of modifications |
|---|---|---|---|
| 1 | The validation lacks strong quantitative benchmarking; the paper does not compare against existing state-of-the-art crowd-risk methods. | We added two quantitative evaluations against ground truth and a baseline. **(a) Counting accuracy vs. ground truth:** the front end now uses benchmark-validated crowd counters; APGCC attains **MAE 48.7 / RMSE 76.7 on ShanghaiTech Part A** (in-domain, ≈ published SOTA) and DM-Count **MAE 14.4 on Part B**, with MOT-ground-truth **MAE 0.55 on PETS2009 S2.L1**; both counters are reported. **(b) Non-linear vs. linear baseline:** on the dense ShanghaiTech-A set, CA-CRS+ flags **55% of images as DANGER while a weight-matched linear density–speed baseline flags 0%** — the linear form misses every high-density crush state, quantifying the contribution of the non-linear Φ term. | §IV-B (Counting validation, new Table); §IV-D (Linear-baseline comparison, new Table) |
| 2 | Several model parameters and coefficients are selected without detailed sensitivity analysis or systematic justification. | We added a sensitivity analysis over all five scoring parameters. The DANGER classification is **robust to the linear weights** ($w_1$ moves the DANGER rate by ⩽10 pts; $w_2$, $w_3$ by 0 pts over their tested ranges) and **sensitive only to the exponential gridlock parameters** (γ: 34-pt range; λ: 48-pt range), which we now report and discuss, motivating careful selection of γ, λ. Speed normalization is additionally grounded in physical camera calibration (see Reviewer 2, item 4). We also soften the "first framework" wording to "to our knowledge, among the first to integrate…". | §IV-E (Sensitivity analysis, new Table/Fig.); §III-C; §I |

# Reviewer 2

| # | Reviewer's comments | Authors' response | Location of modifications |
|---|---|---|---|
| 1 | The primary claimed contribution (ripple-effect prevention) is untested; all DANGER frames are density-dominant, so the REDIRECT pathway and ripple check are never triggered — validate on opposing-flow sequences (PETS2009), do not defer. | We evaluated the conflict pathway on **real bidirectional-flow data (PETS2009 S2.L1)** using a corrected directional-conflict metric (circular variance of flow direction, which — unlike the previous "fraction opposing the dominant direction" — does not degenerate at a balanced opposing split). **CONFLICT is dominant in 66–71% of frames in the crossing-flow view, so REDIRECT is now triggered on real data** (the original evaluation never triggered it). The ripple-downgrade **logic is verified on a controlled two-zone case** (downstream score ⩾ 0.50 ⇒ REDIRECT correctly downgraded to HOLD; < 0.50 ⇒ retained). We state transparently that PETS2009 S2.L1 is sparse (maximum score ≈ 0.50), so **no downstream-crowding condition arises on it and no downgrade is warranted** — the downgrade is therefore demonstrated on the controlled case, not overclaimed on the real sequence. | §IV-C (Conflict + REDIRECT on PETS2009, new Fig.); §III-D (ripple logic); §V (Limitations) |
| 2 | CRR figures are simulation outputs, not empirical measurements, yet presented as primary results; Table III coefficient basis is not adequately justified. | We agree. **All CRR/projected-CRS values are now explicitly labeled simulation estimates conditional on the assumed reduction coefficients (Table III)** — in the abstract, body, every figure, and the Table III caption — and CRR is **removed as a headline result.** We state that the ξ coefficients were not re-derived from pedestrian-flow theory and identify a flow-theoretic derivation as future work; CRR is computed on real motion/count inputs but with assumed multipliers, and is described as such. | Abstract; §III-F; §IV-D; Table III caption; §V |
| 3 | No ground-truth count validation for the occlusion-correction model; person-count accuracy is foundational to all downstream scores. | The person-detection + occlusion-correction front end (which structurally undercounts dense crowds) is **replaced by benchmark-validated crowd counters**, evaluated against ground truth: **APGCC MAE 48.7 (≈SOTA, ShanghaiTech-A), DM-Count MAE 14.4 (Part B), MOT-GT MAE 0.55 (PETS2009 S2.L1).** We disclose a cross-domain undercount bias (mean −10 to −18) and note undercount is the conservative-unsafe direction. | §IV-B (new Table); §III-B; §V |
| 4 | Sub-real-time throughput on the evaluation platform | We now report **measured** end-to-end throughput on a **Tesla T4: 4.0 FPS (0.250 s/frame)** — counter 10.5 FPS, | §IV-F (revised Table V); |

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| | (0.03–0.14 fps); GPU deployment performance entirely estimated. | motion 6.5 FPS, scoring 25,695 FPS. The system is repositioned as a **~1–2 Hz risk-update monitor** (appropriate because crowd density evolves over seconds), and we **no longer claim frame-level real-time**. The previously all-estimated throughput table is replaced with these measured values. | Abstract |
| 5 | Weight/parameter sensitivity analysis absent; five free parameters ($w_1$, $w_2$, $w_3$, $\gamma$, $\lambda$) govern the scoring formula. | Added — see Reviewer 1, item 2. Full five-parameter sensitivity is reported: robust to $w_1$, $w_2$, $w_3$; sensitive to $\gamma$, $\lambda$. | §IV-E (new Table/Fig.) |
| 6 | Scenario D is not a real multi-camera deployment (no shared topology, no spatial calibration, no genuine inter-zone adjacency); state clearly it is a proof-of-concept simulation. | Scenario D is now a **genuine multi-camera evaluation**: three overlapping real camera views (View_001/005/007) of the **same PETS2009 S2.L1 junction**, with camera calibration, fused by robust median. The views genuinely disagree — **mean cross-view count spread 4.18, maximum 11.29** — which the fusion reconciles. We describe it as **real multi-view fusion at the scene's actual (low) density**, explicitly not a crush, and remove any wording implying a physically deployed multi-camera venue. | §III-A; §IV-C (Multi-view, new Fig.); §V |

# Additional revisions (not raised by reviewers, made during revision)

| Area | Change |
|---|---|
| References | Verified all entries against primary sources and added **DOI, issue, page numbers, and month** per the formatting checklist. Corrected: **[2]** (title/volume/page corrected to the actual 2023 *Safety Science* paper), **[9]** (SAFEST venue/year corrected to WISG 2015), **[6]** (Ding volume), **[10]** (title/DOI). |
| Table I | Corrected the **SAFEST "RGB-only" entry** — SAFEST uses distributed **infrared** imaging and is not RGB-only. |
| Equations | Made the **[0,1] saturation explicit** in the score definition; **corrected the GRS urgency weighting to 2:1** (the earlier expression encoded 3:1, contradicting the text). Equation references use the "(1)" form, not "Eq. (1)". |
| Method | Conflict is now **circular variance of flow direction**; speed is **resolution/fps-independent** and normalized by a calibration-derived constant, replacing the 15 px/frame value. |
| Dataset | Retired non-reproducible/uncalibrated stock clips in favor of public, ground-truth benchmarks (PETS2009, ShanghaiTech). |

# Formatting checklist confirmation

Title/abstract free of symbols/math ✔ · No numbers in author names ✔ · IEEE/LaTeX template ✔ · English throughout ✔ · References with page/month/issue/DOI, IEEE style ✔ · "in [n]" reference form ✔ · Figures clear, axis labels with units in parentheses ✔ · Figures/tables at column top/bottom ✔ · Equations as "(n)" ✔